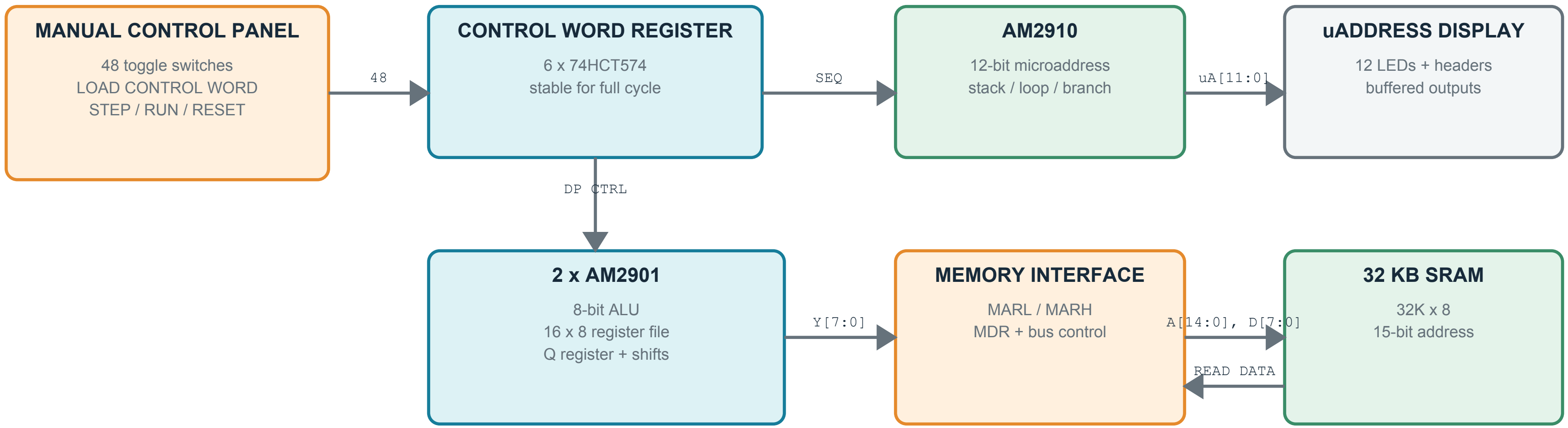
AM2901 / AM2910
Evaluation and Learning Board

First architectural draft: 8-bit datapath, manual microcontrol, visible state, and 32 KB SRAM

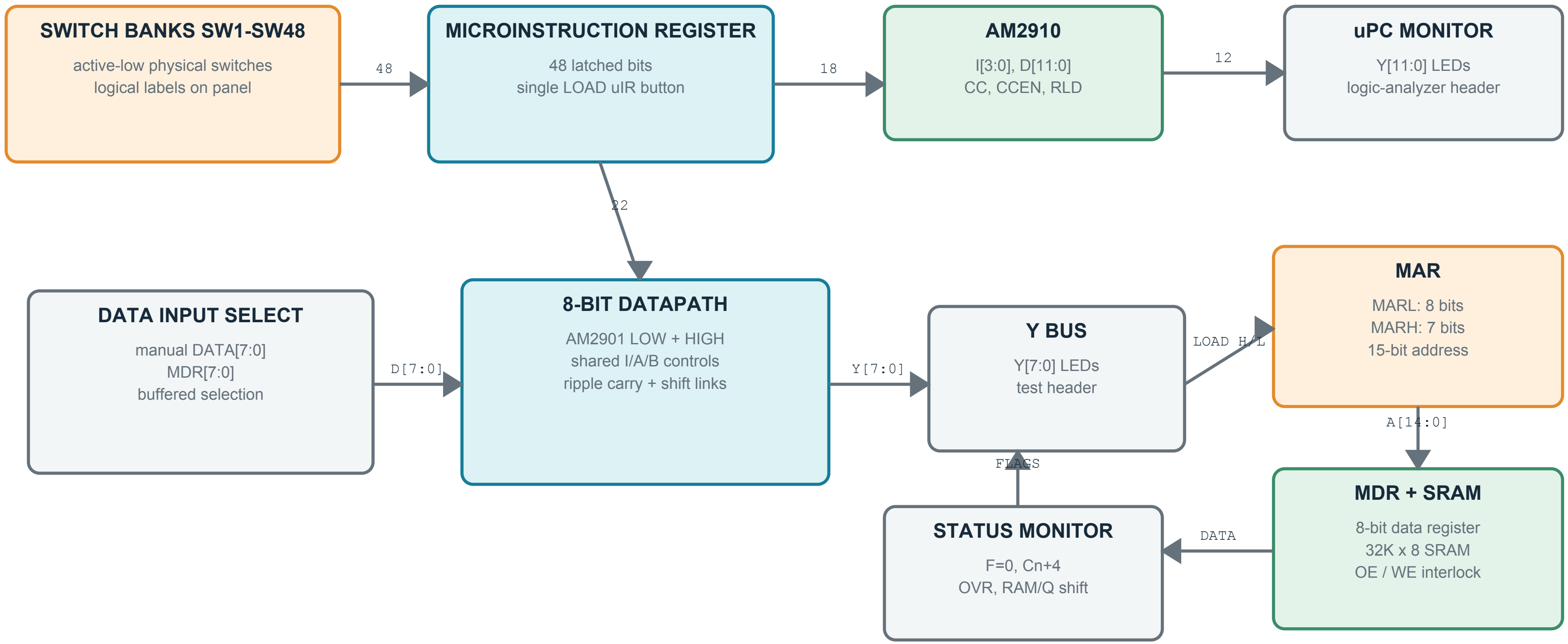
DRAFT 0.1 | THROUGH-HOLE | 5 V LOGIC | MANUAL FIRST



Purpose	Reference	Boundary
A bench-top trainer for learning the AM2901 datapath and AM2910 sequencing one microcycle at a time.	Adapted from the local AMD Am2900 Evaluation and Learning Kit manual, while replacing its controller with an AM2910 and expanding the datapath and memory.	This PDF is an architecture and control-panel draft. It is not yet a manufacturing schematic or PCB release.

1. Functional Architecture

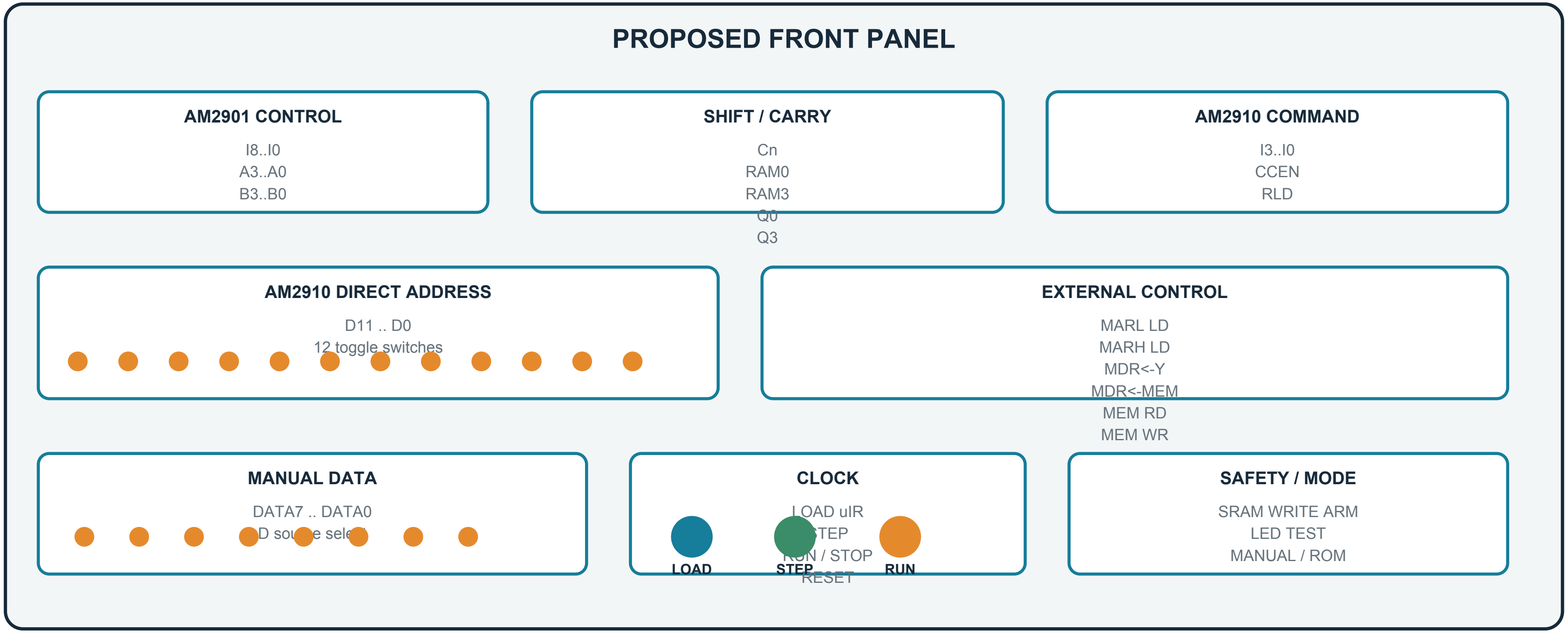
The board is deliberately split into observable modules so each section can be tested independently.



Manual mode Switch settings are captured into a 48-bit microinstruction register. Pressing STEP clocks the selected AM2901 and AM2910 actions.	Sequencer mode The AM2910 calculates and displays the next 12-bit microaddress even before a control-store daughterboard is installed.	Memory mode The 8-bit Y bus can load the low or high address byte and the MDR, allowing explicit SRAM read/write experiments.

2. Manual Control and Monitoring

A physical panel makes every micro-operation intentional and visible.

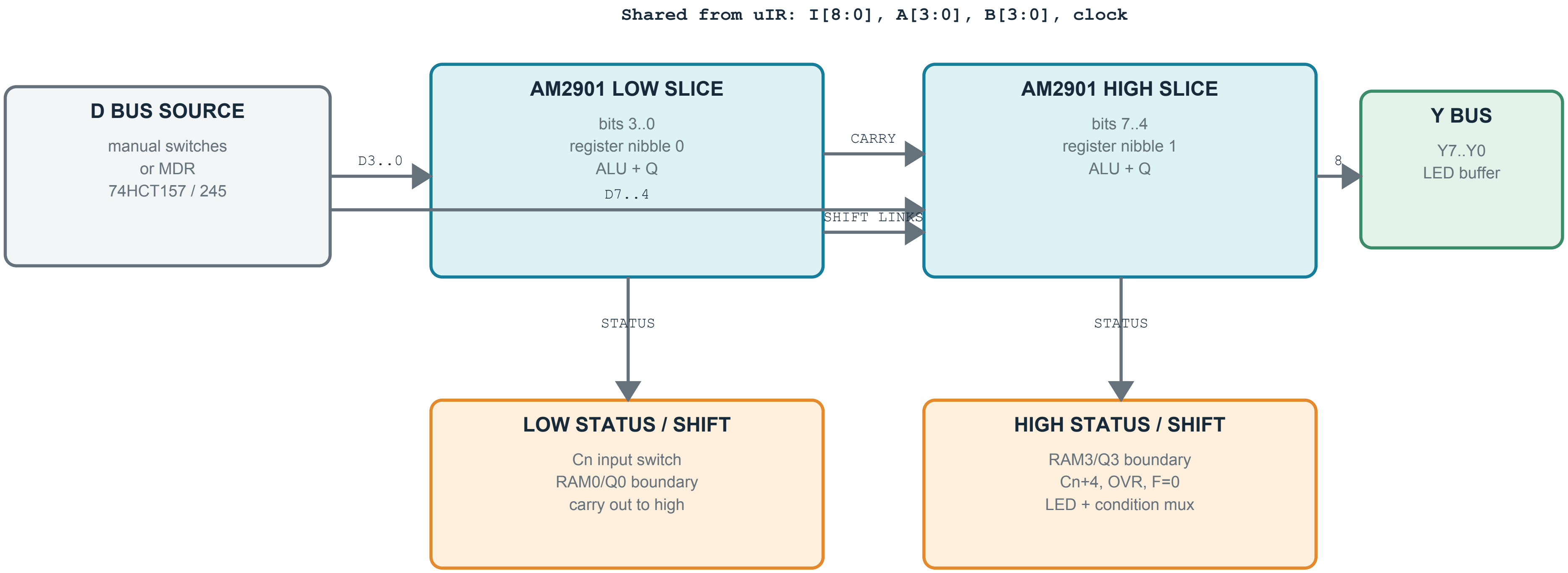


Interaction model: set switches, press LOAD uIR, then press STEP. The SRAM write control should require both a microcontrol bit and a separate guarded WRITE ARM switch.

LEDs must be driven through buffers or transistor arrays. Do not connect a large LED bank directly to the AMD outputs.

3. AM2901 Datapath

Two 4-bit slices operate in parallel as one 8-bit ALU and 16 x 8 register file.

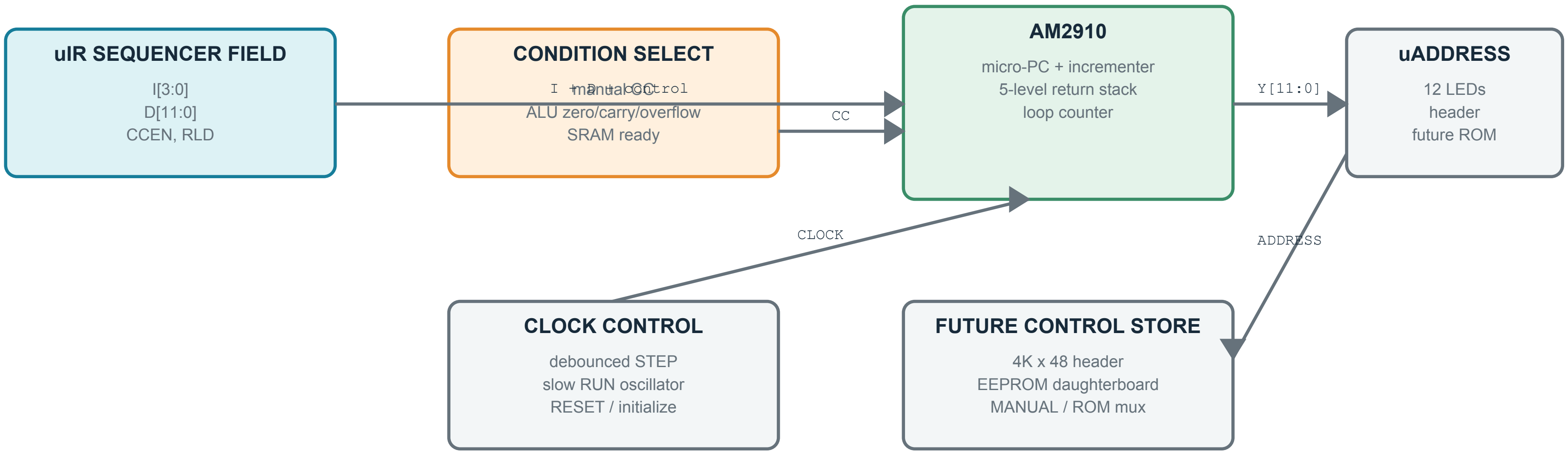


Shared controls Both slices receive identical I[8:0], A[3:0], B[3:0], and clock signals.	Nibble wiring The low AM2901 connects to D/Y bits 3..0; the high AM2901 connects to bits 7..4.	Inter-slice links Carry and shift boundary pins are chained. Exact polarity and pin numbers must be copied from the selected AM2901B datasheet during schematic capture.

First bring-up sequence: load two AM2901 registers, read each through Y, add them, test carry with FF + 01, then test Q and RAM shifts across the nibble boundary.

4. AM2910 Sequencer and Control Word

The sequencer is useful immediately as a visible next-address machine; a ROM board can be added later.

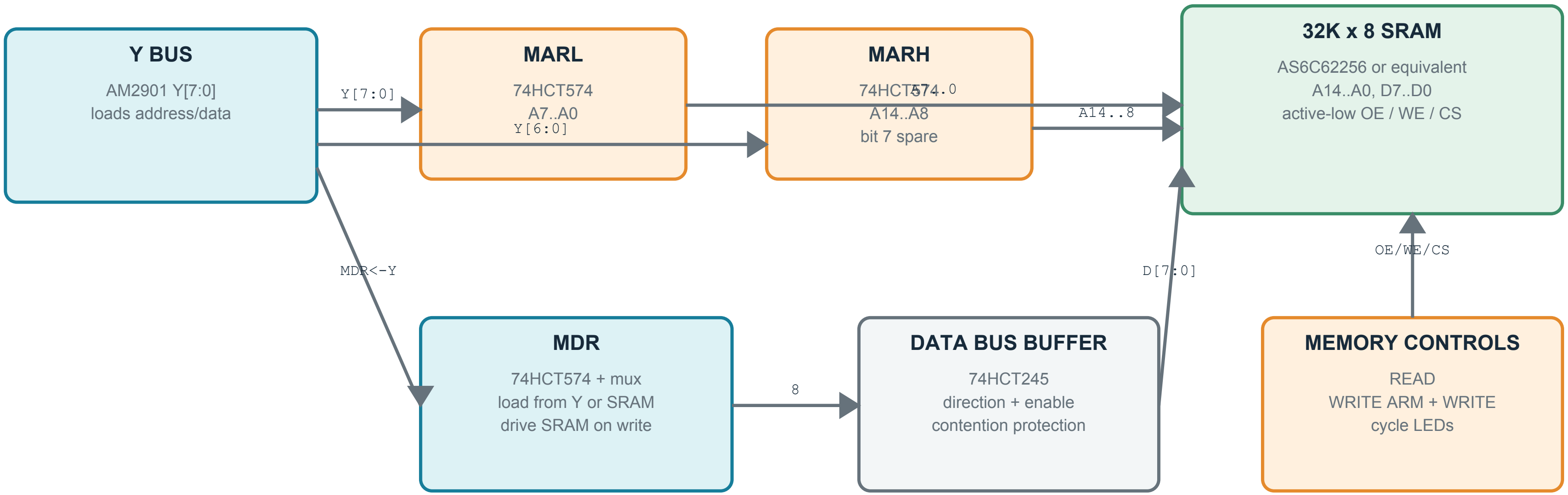


Bits	Field	Width	Destination / purpose
47..39	ALU_I	9	AM2901 I[8:0]: source, function and destination
38..35	A_ADDR	4	AM2901 A-port register address
34..31	B_ADDR	4	AM2901 B-port register address
30..26	SHIFT_CARRY	5	Cn, RAM0, RAM3, Q0, Q3
25..22	SEQ_I	4	AM2910 instruction inputs
21..10	SEQ_D	12	AM2910 direct/branch address
9	CCEN	1	Enable selected condition into sequencer
8	RLD	1	AM2910 loop-counter load control
7..0	EXT	8	MARL/H load, MDR paths, memory read/write, spare

The 48-bit allocation is intentionally direct and roomy. The final EXT bit assignment may change after the pin-level schematic exposes clock-edge, output-enable, and memory-cycle requirements.

5. 32 KB SRAM Subsystem

A 15-bit memory address is built in two visible byte-sized steps.



Experiment	Manual micro-operations
Set address	Place low byte on Y and load MARL; place high seven bits on Y and load MARH.
Read byte	Assert SRAM read, capture SRAM data into MDR, then select MDR as the AM2901 D input.
Write byte	Load MDR from Y, enable WRITE ARM, then assert the SRAM write microcontrol for one controlled cycle.

The SRAM output enable and write enable must be interlocked so they cannot be asserted together. All bus drivers should default disabled during reset.

6. Preliminary Parts List

Draft choices favor socketed, through-hole, 5 V-compatible parts and easy probing.

Qty	Part / function	Draft choice	Notes
2	4-bit bit-slice processor	AM2901B	Low and high nibbles; socketed
1	Microprogram sequencer	AM2910APC	12-bit address; socketed
1	32K x 8 SRAM	AS6C62256-55PCN	5 V-capable DIP-28 candidate; verify chosen vendor
6	Microinstruction register	74HCT574	Latches 48 manual control bits
2	Memory address registers	74HCT574	MARL and MARH; one high bit unused
1-2	Memory data register / path	74HCT574 + 74HCT157	Select Y or SRAM as MDR input
2	Bus transceiver / isolation	74HCT245	Prevent contention on D and SRAM buses
4-6	LED output buffers	74HCT541 / ULN2803A	Final count depends on displayed signals
6	8-way switch bank	DIP or toggle bank	48 microcontrol switches
1	8-way data switch bank	DIP or toggle bank	Manual D input
2	Debounced clock channels	74HCT14 + 74HCT74	STEP and slow RUN
1	Condition multiplexer	74HCT151	Select ALU/manual/memory condition
1	Reset supervisor	MCP100 / RC + Schmitt	Choice to be finalized
many	LEDs, resistor networks, headers	2.54 mm through-hole	Logic-analyzer access encouraged

Power
Regulated 5 V input, reverse-polarity protection, bulk capacitance per board region, and one 100 nF ceramic capacitor at every IC.

Grounding
Use a solid ground plane on the eventual PCB. Keep clock, SRAM write, and reset routing short and well separated from LED currents.

Current budget
Original bipolar AMD parts and many LEDs can draw substantial current. Size the regulator only after checking the exact device suffixes.

7. Build Stages and Open Decisions

The design should be proven in layers before committing to a large PCB.

Stage	Hardware fitted	Acceptance test
A - clock and panel	Clock/reset, switch latches, LED buffers	Every latched switch and clock mode is observable and bounce-free.
B - 8-bit datapath	Two AM2901s and D/Y bus	Register load/read, ALU functions, carry, Q and cross-slice shifts pass.
C - sequencer	AM2910 and condition mux	Continue, jump, conditional branch, call/return, and loop operations pass.
D - memory	MARL, MARH, MDR, bus buffers, SRAM	Walking-bit address/data tests pass across all 32 KB.
E - control store	Optional 4K x 48 EEPROM daughterboard	Manual/ROM selection and automatic microprogram execution pass.

Decide before KiCad pin wiring	Recommended default for draft 0.2
Exact AM2901 and AM2910 manufacturer/suffix; preferred switch style; board versus separate front panel; SRAM part; LED color/count; onboard control-store sockets.	Use the local AM2901B and AM2910APC datasheets, through-hole toggles or DIP switches, a ribbon-connected front panel, AS6C62256 SRAM, and a 4K x 48 expansion header.

Local references used: `EVAL_Board_cropped_reduced.pdf`, `AM2901B.PDF`, and `AMD-AM2910APC-datasheet.pdf` in the Hardware Projects/AM29xx folder.

Next engineering artifact: hierarchical KiCad schematic with verified pin numbers, active-low signal names, power pins, test points, and ERC annotations.